

Identifying and Predicting the Effect of Sunspot Numbers on Global Temperatures and Climate Change



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Abstract:

The relationship between solar activity, as indicated by historical sunspot numbers, and global temperature variations remains a topic of ongoing scientific investigation. In our study, we attempt to draw parallels between solar activity and observed climate patterns.

Using our primary methodology—wavelet analysis—we examined historical sunspot data and global temperature records and isolated frequencies to understand their correlation. Then, we inputted the significant outputs from the wavelet analysis into a neural network to predict future cycles of Long Island (Fig. 2) temperature and sunspots. Additionally, our research attempted to rule out confounding variables such as human-induced factors in amplifying recent climate change; in turn, this allowed us to determine if there was a causal relationship between natural solar influences and temperature.

By analyzing the 11-year solar cycle through the wavelet power spectrum, the study found that periodic fluctuations in global temperatures corresponded to peaks in solar activity. However, the findings of our research suggest that while solar activity has historically influenced temperature fluctuations, human activity instead is the dominant force driving recent climate changes.

For future research, understanding these interactions between solar cycles and global temperatures is crucial for improving climate models and predicting future climate trends. This knowledge could help mitigate the effects of climate change, particularly in regions vulnerable to temperature-related environmental changes.

Background Information:

- There has been a significant increase in global temperatures in the past 50 years—this is due to a phenomenon scientists call “global warming” (Li Z.).
- While much of the temperature increase is attributed to human activity, there is historical evidence (namely the “Little Ice Age” & “Medieval Climatic Optimum”) suggesting that solar activity may play a crucial role in temperature fluctuations (Eddy J. A.).
- Sunspots are darker regions on the photosphere where the Sun’s magnetic flux is much greater than in the surrounding areas.
- The solar cycle describes a pattern of changing sunspot numbers over a period of 11 years (Fig. 1) (NOAA).

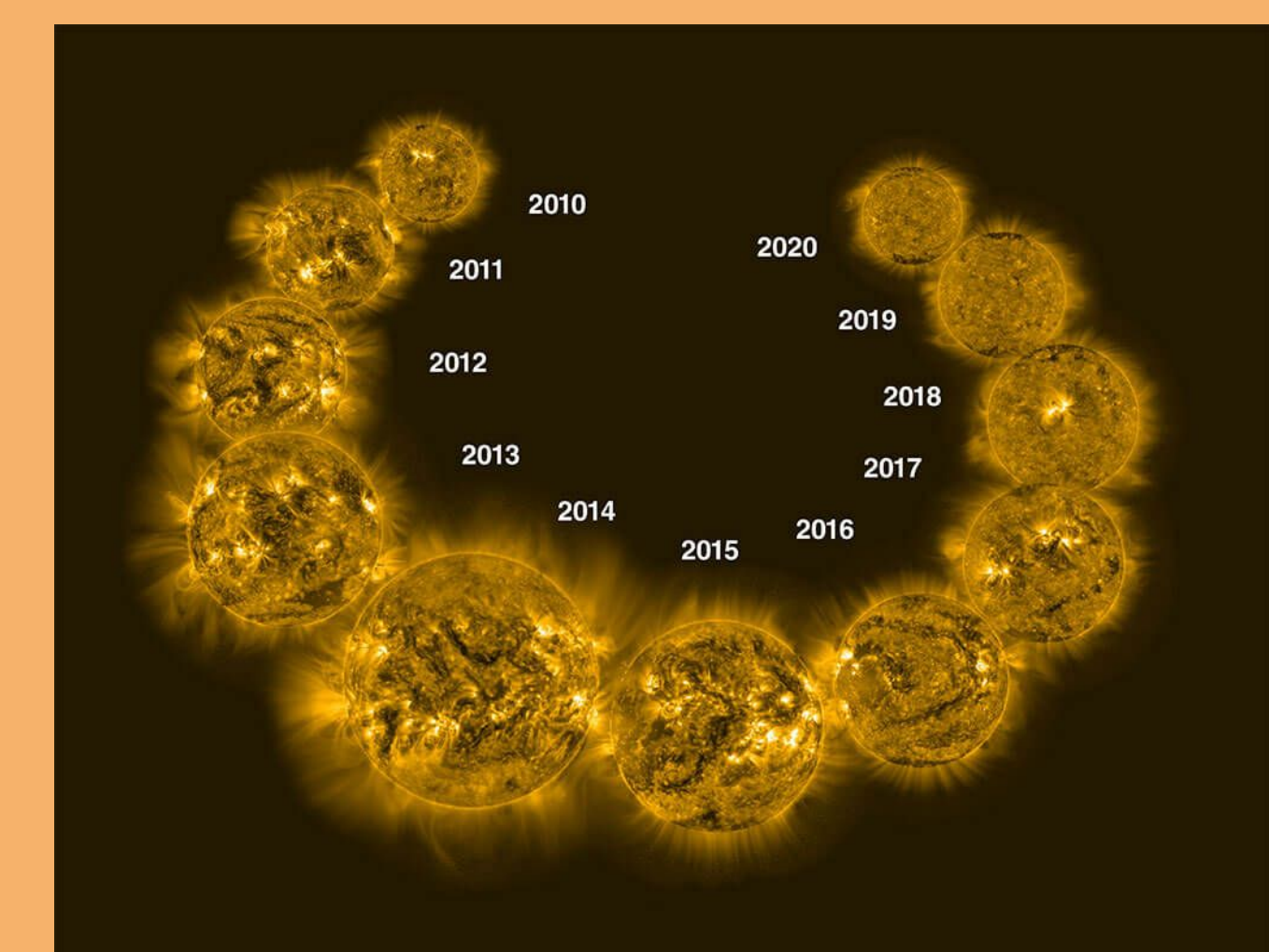


Figure 1: Conceptual figure showing the fluctuation/orientation of sunspots throughout the 11 year solar cycle.

Method/Materials:

- Sunspot Dataset**
 - Downloaded into RStudio through .csv file
 - http://www.sidc.be/silso/DATA/SN_d_tot_V2.0.csv
- US National Centers for Environmental Information**
 - Atmospheric data from 1948-2024
 - Location: Long Island, New York, USA
- Google Maps**
 - Creation of a map of Long Island
- RStudio (v.4.2.1)**
 - Perform Wavelet Analysis
 - Train and Test Neural Network
 - Packages used: WaveletComp, neuralnet, etc.
 - Moving average (kza)
 - Temperature data package (GSODR)

Results:

- Wavelet analysis was used to explore patterns in historical temperature data and sunspot numbers. The wavelet power spectrum for temperatures revealed a strong annual cycle, highlighting the regular seasonal variations each year (Fig. 5).
- The sunspot data showed a significant 11-year cycle, which aligns with the usual cycle of solar activity (Fig. 6).
- These findings provide a basis for examining how fluctuations in solar activity, as indicated by sunspot cycles, might have influenced global temperature trends and climate change over time.
- Our temperature neural network predicted a consistent seasonal temperature change occurring yearly (Fig. 4).
- Our sunspot neural network predicted that sunspot numbers will experience a significant increase about 5 years into the future and then decrease (Fig. 3)

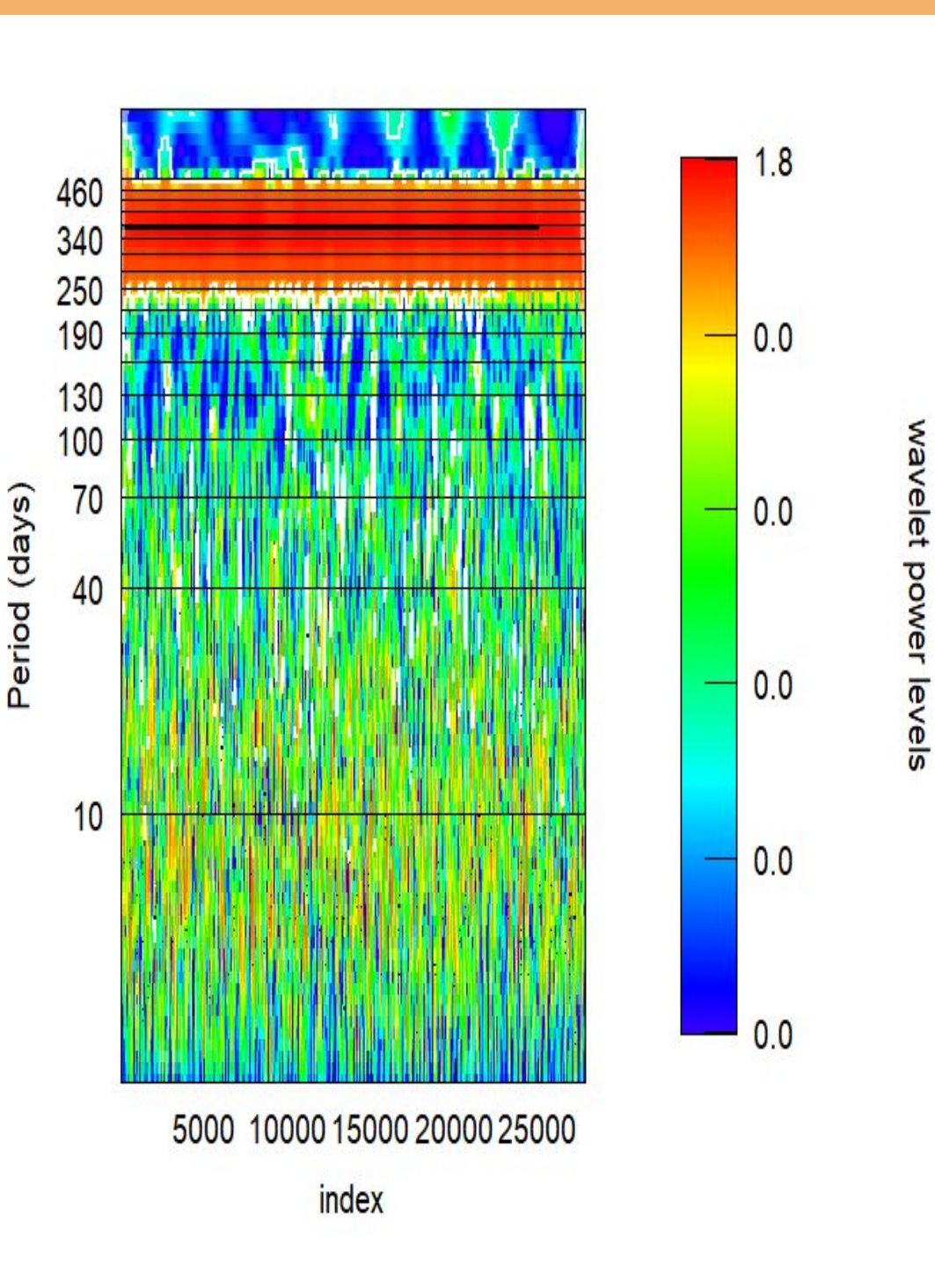


Figure 5: Temperature power spectrum plot

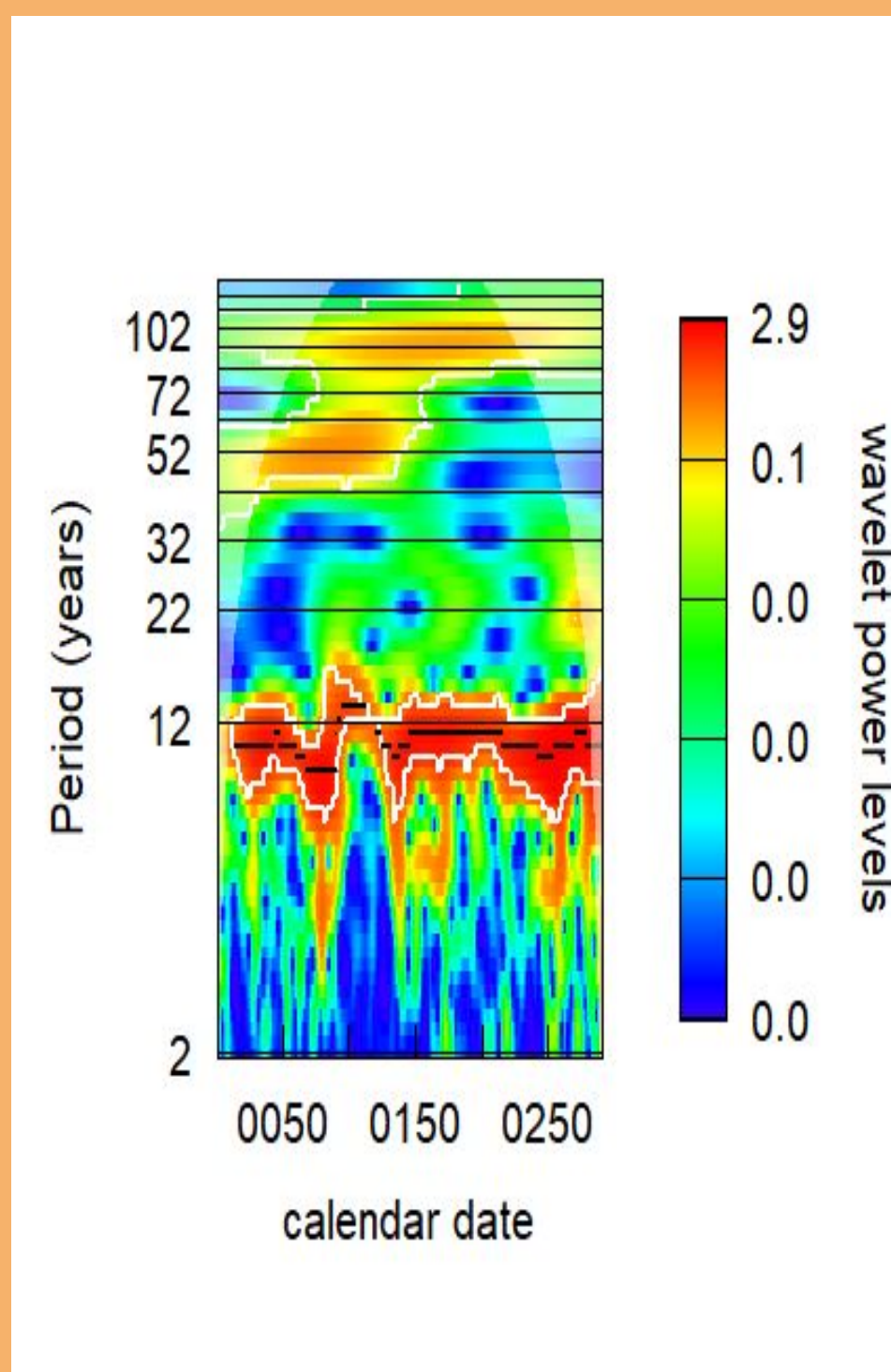


Figure 6: Sunspot power spectrum plot



Figure 2: Map of Long Island from Google Maps.

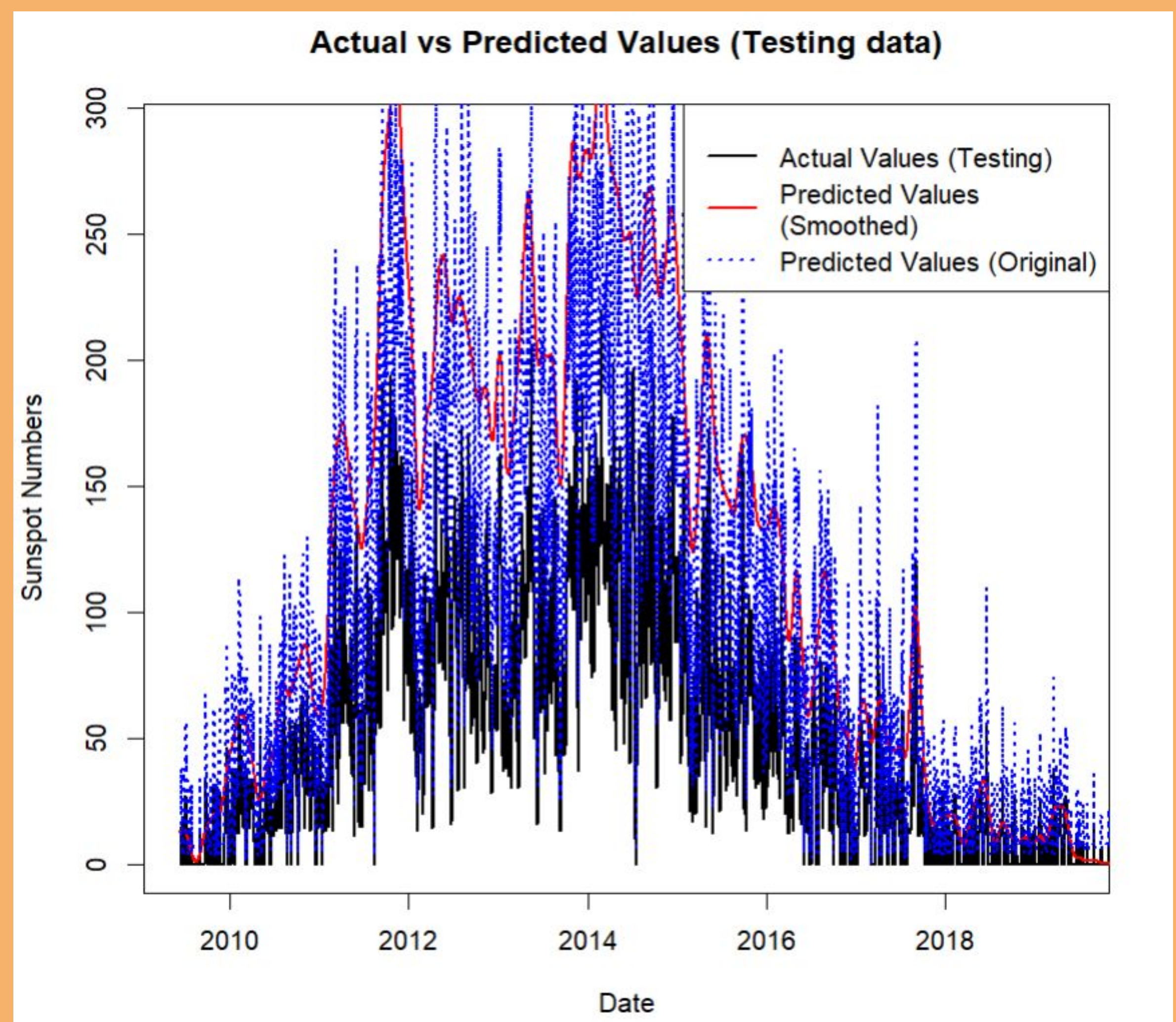


Figure 3: Neural network prediction of sunspot numbers up to 10 years into the future. The KZ-smoothed predicted values are represented in red.

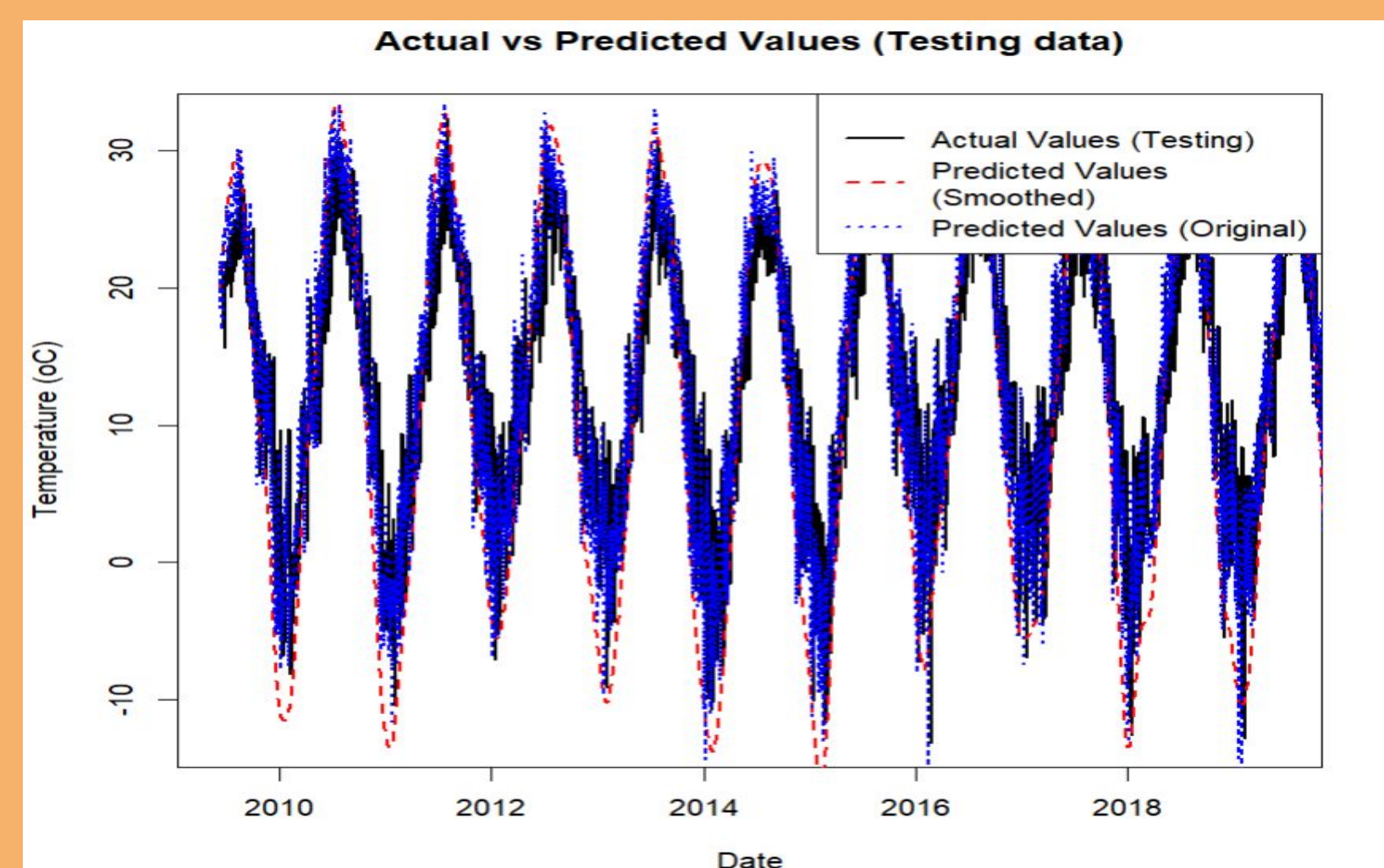


Figure 4: Neural network prediction of Long Island temperatures up to 10 years in the future.

Discussion/Conclusion:

- Strong annual frequency for temperatures
- Strong 11 year frequency for sunspots
- These findings coincide with established research and our neural network predictions.
- The frequencies themselves do not show a powerful relationship or similarity.
- While solar cycle and temperatures may be correlated, more experimentation needs to be done to solidify a cause/effect relationship.

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Credit Authorship Contribution Statement:

Lolla, H.: editing, literature, Rstudio coding, Figure 4, Figure 5, writing- Results, Discussion; Sharma, E.: Literature, editing, writing- Global Mapper, Google Earth, Abstract, Results- figure 1, Methods, Discussion; Lin, A.: references, literature, editing, writing - Introduction, Discussion.

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